1.

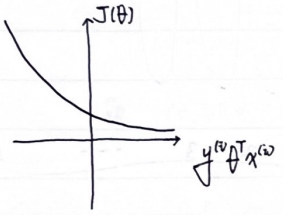
(a) B didn't converge.

(b) B is linearly seperable (on plot)

A is not.

$$\nabla_\theta J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} \frac{y^{(i)}}{1+\exp(y^{(i)}\theta^T x^{(i)})} x^{(i)}$$

$$\therefore \int_\theta \nabla_\theta J(\theta) = J(\theta) = -\frac{1}{m}\sum_{i=1}^{m} \log\left(\frac{1}{1+\exp(-y^{(i)}\theta^T x^{(i)})}\right)$$



데이터가 완전히 linearly seperable 하다면,

$\forall i,\ y^{(i)}\theta^T x^{(i)} > 0$ (functional margin)

즉, 모든 case에 대해서 올바르게 분류가 가능하다.

그런데, 이때도 θ를 크게 스케일링해서 무한정 $J(\theta)$를 작게할수 있음.

$J(\theta) \downarrow$, $\Longrightarrow$ not converge!

(c)

(i) No

(ii) Yes.

(iii) No.

(iv) Yes.

(v) Yes.

(d) hinge loss : $J(\hat{y}) = \max(0, 1-y\cdot\hat{y})$ where $\hat{y} = w^T x + b$.

In linearly seperable case, $y\cdot\hat{y} > 0$. for all.

and by scaling, we can make $|\hat{y}| \geq 1$, so $J(\hat{y}) = 0$.

$\Rightarrow$ it can be converged. so. SVM with hinge loss isn't vulnerable. to this case.

2.

(a)

we choose θ

such that $\theta = \arg\max_\theta l(\theta) = \arg\max_\theta \sum_{i=1}^{m} \log h_\theta(x^{(i)})^{y^{(i)}} (1-h_\theta(x^{(i)}))^{1-y^{(i)}}$

$\nabla_\theta l(\theta) = \sum_{i=1}^{m} (y^{(i)} - h_\theta(x^{(i)})) x^{(i)}$

θ^* satisfies $\sum_{i=1}^{m} (y^{(i)} - h_\theta(x^{(i)})) x^{(i)} = 0.$

$x_0^{(i)}$ (Interceptor) $= 1$. so, $\sum_{i=1}^{m} (y^{(i)} - h_{\theta^*}(x^{(i)})) = 0.$

$\therefore \sum_{i=1}^{m} y^{(i)} = \sum_{i=1}^{m} h_{\theta^*}(x^{(i)})$

$\therefore \frac{\sum_{i \in I_{a,b}} p(y^{(i)}=1 \mid x^{(i)};\theta)}{|\{i \in I_{a,b}\}|} = \frac{\sum_{i=1}^{m} h_\theta(x)}{m} = \frac{\sum_{i=1}^{m} y^{(i)}}{m} = \frac{\sum_{i \in I_{a,b}} I\{y^{(i)}=1\}}{|\{i \in I_{a,b}\}|}$

if $(a,b) = (0,1)$

(b) ~~perfectly~~ calibrated $\Longleftrightarrow$ perfect accuracy ?

(i) ($\Longrightarrow$) no !

$$\frac{\sum_{i \in I_{a,b}} I\{y^{(i)}=1\}}{|\{i \in I_{a,b}\}|} = \frac{\sum_{i \in I_{a,b}} p(y^{(i)}=1 \mid x^{(i)};\theta)}{|\{i \in I_{a,b}\}|}$$

if $(a,b) = (0.5, 1)$ $\quad \frac{\sum_i p(y^{(i)}=1 \mid x^{(i)};\theta)}{|\{I_{a,b}\}|} = \frac{\sum h_\theta(x^{(i)})}{|\{I_{a,b}\}|} < 1.$

$\therefore \frac{\sum_i I\{y^{(i)}=1\}}{|\{I_{a,b}\}|} < 1.$ in $(0.5, 1)$ which means that not perfect accuracy

(ii) ($\Longleftarrow$) no !

if perfect accuracy, in $(0.5, 1)$

$$\frac{\sum_i I\{y^{(i)}=1\}}{|\{I_{a,b}\}|} = 1 > \frac{\sum_{i \in I_{a,b}} h_\theta(x^{(i)})}{|\{I_{a,b}\}|}$$

So not perfectly calibrated.

(c) L_2 Regularization:

$$\nabla_\theta \ell(\theta) = \sum_{i=1}^{m} (y^{(i)} - h_\theta(x^{(i)}))\, x^{(i)} - 2\lambda\theta$$

$\because x_0 = 1$, so,

$$\sum_{i=1}^{m} (y^{(i)} - h_\theta(x^{(i)})) - \lambda\theta_0 = 0.$$

$$\therefore \sum h_\theta(x^{(i)}) + \lambda\theta_0 = \sum y^{(i)}$$

So, if $\theta_0 = 0$, calibration still hold, otherwise not.

3.

(a)

$$p(\theta|x,y) = \frac{p(x,y|\theta)p(\theta)}{p(x,y)} = \frac{p(x,y,\theta)}{p(x,y)} = \frac{p(y|x,\theta)p(x,\theta)}{p(x,y)}$$

$$= \frac{p(y|x,\theta)\,p(\theta|x)\,p(x)}{p(x,y)}$$

$$\therefore p(\theta|x,y) = \frac{p(x)}{p(x,y)}\, p(y|x,\theta)\, p(\theta|x). \quad (\text{if } p(\theta|x) = p(\theta))$$

$$= \frac{p(x)}{p(x,y)}\, p(y|x,\theta)\, p(\theta).$$

$$\therefore \arg\max_\theta p(\theta|x,y) = \arg\max_\theta p(y|x,\theta)\, p(\theta).$$

(b)

$$\arg\min_\theta -\ell(\theta) = \arg\min_\theta -\log p(\theta|x,y) = \arg\min_\theta -\log p(y|x,\theta) - \log p(\theta)$$

$$= \arg\min_\theta -\log p(y|x,\theta) - \log \frac{1}{(2\pi)^{p/2}|\eta^2 I|^{1/2}} \exp\left(-\frac{1}{2}\theta^T[\eta^2 I]^{-1}\theta\right)$$

$$= \arg\min_\theta -\log p(y|x,\theta) + \frac{1}{2\eta^2}\|\theta\|_2^2 \quad \left(\because \lambda = \frac{1}{2\eta^2}\right)$$

(c)

$y|x;\theta \sim \mathcal{N}(\theta^T x, \sigma^2)$, $\theta \sim \mathcal{N}(0, \eta^2 I)$

$$\arg\min_\theta -\log \prod_{i=1}^m p(y^{(i)}|x^{(i)};\theta) + \frac{1}{2\eta^2}\|\theta\|_2^2$$

$$= \arg\min_\theta \sum_{i=1}^m \frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} + \frac{1}{2\eta^2}\|\theta\|^2$$

$$\hookrightarrow J(\theta) = \frac{1}{2\sigma^2}(X\theta - \vec{y})^T(X\theta - \vec{y}) + \frac{1}{2\eta^2}\|\theta\|_2^2$$

$$\nabla_\theta J(\theta) = \frac{1}{\sigma^2}(X^T X\theta - X^T y) + \frac{1}{\eta^2}\theta \overset{\text{set}}{=} \vec{0}$$

$$\therefore \theta_{map} = (X^T X + \frac{\sigma^2}{\eta^2} I)^{-1} X^T y$$

(d)

$$\log p(\theta) = \log \prod_{i=1}^n \frac{1}{2b} \exp\left(-\frac{|\theta|}{b}\right)$$

$$= \log\left(\frac{1}{2b}\right)^n + \sum_{i=1}^n \left(-\frac{|\theta|}{b}\right)$$

$$= \log\left(\frac{1}{2b}\right)^n - \frac{1}{b}\|\theta\|_1$$

$$J(\theta) = \arg\min_\theta \frac{1}{2\sigma^2}(X\theta - \vec{y})^T(X\theta - \vec{y}) + \frac{1}{b}\|\theta\|_1$$

$$\therefore \gamma = \frac{2\sigma^2}{b}$$

4.

(a) $z^T k z = z^T(k_1 + k_2) z = z^T k_1 z + z^T k_2 z \geq 0$. So $k \succeq 0$. (o)

(b) $z^T k_1 z - z^T k_2 z$ is not always PSD if $k_2 = 2k_1$ (x)

(c) $z^T a k_1 z = a z^T k_1 z \geq 0$ (o)

(d) $-a z^T k_1 z \leq 0$. (x)

(e) $z^T k z = \sum_i \sum_j z_i k_{ij} z_j = \sum_i \sum_j z_i k_1(x^{(i)}, x^{(j)}) k_2(x^{(i)}, x^{(j)}) z_j = \sum_i \sum_j z_i \phi_1(x^{(i)})^T \phi_1(x^{(j)}) \phi_2(x^{(i)})^T \phi_2(x^{(j)}) z_j$

$$= \sum_i \sum_j z_i \sum_k \phi_{1k}(x^{(i)}) \phi_{1k}(x^{(j)}) \sum_l \phi_{2l}(x^{(i)}) \phi_{2l}(x^{(j)}) z_j$$

$$= \sum_k \sum_l \sum_i \sum_j z_i \phi_{1k}(x^{(i)}) \phi_{2l}(x^{(i)}) z_j \phi_{1k}(x^{(j)}) \phi_{2l}(x^{(j)}) = \sum_k \sum_l \left(\sum_i z_i \phi_{1k}(x^{(i)}) \phi_{2l}(x^{(i)})\right)^2 \geq 0.$$

(o)

(f) $z^T K z = z^T f(x) f(x)^T z = (z^T f(x))^2 \geq 0$ (o)

(g) K is kernel because K_3 is kernel (o)

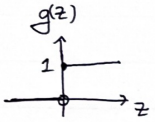
(h) $p(K_1(x,z)) = \sum_k a_k K_1(x,z)^k$

$K_1(x,z)^k$ is kernel ($\because$ (e)) $a_k K_1(x,z)^k$ is kernel ($\because$ (c))

$\sum_k a_k K_1(x,z)^k$ is kernel ($\because$ (a)) (o)

5

(a)



$\theta^{(0)} = 0.$

$\theta^{(1)} = \alpha(y^{(1)} - g(\theta^{(0)T} x^{(1)}))\, x^{(1)}$

$\theta^{(2)} = \alpha(y^{(1)} - g(\theta^{(0)T} x^{(1)}))\, x^{(1)} + \alpha(y^{(2)} - g(\theta^{(1)T} x^{(2)}))\, x^{(2)}$

$\vdots$

$\therefore \theta^{(i)} = \sum_{j=1}^{i} \alpha(y^{(j)} - g(\theta^{(j-1)T} x^{(j)}))\, x^{(j)}$

$= \sum_{j=1}^{i} \alpha(y^{(j)} - g(\theta^{(j-1)T} \phi(x^{(j)})))\, \phi(x^{(j)})$

$h_{\theta^{(i)}}(x^{(i+1)}) = g(\theta^{(i)T} x^{(i+1)})$

$= g(\sum_{j=1}^{i} \alpha(y^{(j)} - g(\theta^{(j-1)T} \phi(x^{(j)})))\, \phi(x^{(j)})^T \cdot \phi(x^{(i+1)}))$

$= g(\sum_{j=1}^{i} \alpha(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}))\, k(x^{(j)}, x^{(i+1)}))$

$\theta^{(i+1)} = \sum_{j=1}^{i+1} \alpha(y^{(j)} - h_{\theta^{(j-1)}}(x^{(j)}))\, \phi(x^{(j)})$

$\theta^{(0)} = 0. \iff h_{\theta^{(0)}}(x^{(1)}) = 1.$

$\theta^{(1)} = \alpha(y^{(1)} - h_{\theta^{(0)}}(x^{(1)}))\, \phi(x^{(1)}) \iff h_{\theta^{(1)}}(x^{(2)}) = \alpha(y^{(1)} - h_{\theta^{(0)}}(x^{(1)}))\, k$

$\theta^{(2)} = \theta^{(1)} + \alpha(y^{(2)} - h_{\theta^{(1)}}(x^{(2)}))\, \phi(x^{(2)}) \iff h_{\theta^{(2)}}(x^{(3)}) = \alpha(y^{(1)} - h_{\theta^{(0)}}(x^{(1)}))\, k + \alpha(y^{(2)} - h_{\theta^{(1)}}(x^{(2)}))\, k$